

Hifz Companion

The method behind the app

How the classical three-fold ḥifẓ discipline — sabaq, sabqi, manzil — is joined to the evidence on spaced repetition and retrieval practice, and turned into a scheduler.

mazhar17.github.io/hc

Version 1.21.4 · free, no account, works offline

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Every research claim in this deck carries its source; the full reference list is on the second-last slide.

THE PROBLEM

Memorising is not the hard part

A page can be committed to memory in a week. Holding six hundred of them for a lifetime is a different problem — and it is a scheduling problem.

“Commit yourself to the Qur’an, for by Him in whose Hand is my soul, it is surely more prone to break away than a camel in its bind.”

Ṣaḥīḥ al-Bukhārī 5033 (Book 66, Hadith 56)

The mudārasah precedent

Jibrīl would review the Qur’an with the Prophet ﷺ once each year, and twice in the year he died.

Ṣaḥīḥ al-Bukhārī 4997–4998.

So the app never asks “have you read this page?” It asks: which pages must be recited from memory today, and how did each one go?

Hadith texts as translated on sunnah.com; the app links sources rather than reproducing them.

The classical frame: three kinds of work

The madrasa curriculum has long divided a ḥāfiẓ's daily work into three layers. The app schedules all three; it does not invent them.

1

Sabaq — the new lesson

The page being memorised now.

One page a week by default; a half, third or quarter page is equally valid.

Across the seven days: learn segment by segment, join, then consolidate.

2

Sabqi — recent revision

Pages learned in the last weeks — still fragile.

Revised at growing intervals until they hold.

This is the layer where ḥifẓ quietly fails, so the app gives it priority.

3

Manzil — the rotation

Everything older, on a steady cycle (30 days by default).

A page is admitted only after it has proved itself — three good recitations at ≥ 30 -day spacing.

Terminology varies by region: in some traditions the recent layer is called sabqi, in others dawr or murāja'ah. The app uses the South-Asian naming.

Where the three terms come from

- They are pedagogical vocabulary, not Qur’anic or hadith terms. No verse or hadith names sabaq, sabqi or manzil as a study scheme.
- Sabaq (سبق) — “lesson”; in Persian and Urdu madrasa usage, the day’s new portion.
- Sabqi (سبقی) — “belonging to the lesson”; the recently-learned portion still under close revision.
- Manzil (منزل) — “stage / stopping-place”. Two distinct senses: (a) one of the seven traditional divisions of the Qur’an for a weekly khatm, and (b) in ḥifẓ usage, the consolidated older portion in long rotation. The app uses sense (b).

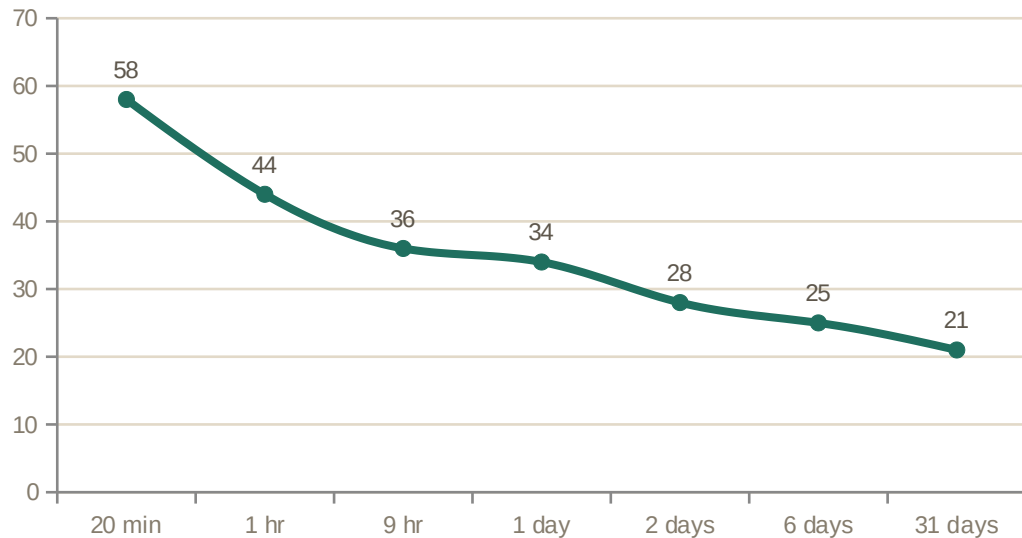
What is attested

- The seven-manzil division of the Qur’an is long attested in the tradition.
- The three-layer daily routine is attested as widespread madrasa practice, transmitted through teachers.
- No classical treatise fixes the interval lengths. Those are the app’s own choice.

Stated honestly: this deck cites the practice as living pedagogy, not as a ruling. The interval arithmetic is modern and evidence-based, not traditional.

Why any of this needs a schedule

Ebbinghaus (1885) measured his own retention of learned material over time. Recall falls steeply at first, then flattens — and each successful review flattens the curve further.



What the app takes from this

Not the numbers — Ebbinghaus used nonsense syllables on a single subject; Qur'anic text is meaningful and heard aloud.

What transfers is the shape: forgetting is fastest right after learning, so the first review must come soon and later ones can be far apart.

That is the shape of the ladder.

Ebbinghaus, H. (1885/1913). Memory: A Contribution to Experimental Psychology. Figures are his savings-method percentages.

Principle 1 — spacing beats massing

Cepeda, Pashler, Vul, Wixted & Rohrer (2006) — a meta-analysis of a century of distributed-practice research.

184

articles reviewed

317

experiments

839

assessments of the spacing effect

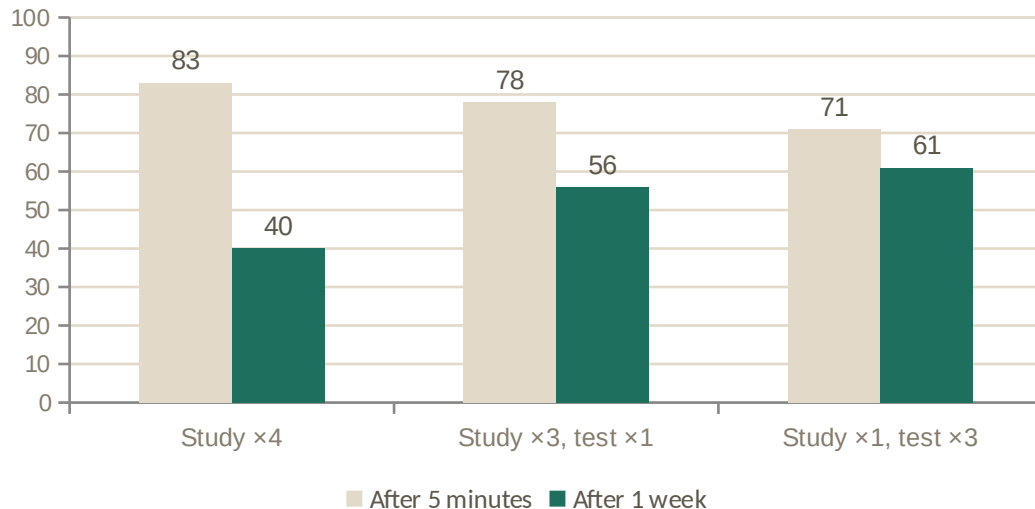
The finding, and the part the app depends on

Spaced practice reliably outperformed massed practice. More importantly for a scheduler: the best gap between reviews is not fixed — it grows with how long you need to remember the material.

Hiŕz is the extreme case: the target is the rest of your life, so the app's intervals climb into the tens of days and stay there.

Principle 2 — recall, don't re-read

Roediger & Karpicke (2006): students who were tested on a passage remembered far more a week later than students who re-read it — even though re-reading felt better at the time.



In the app: the curtain

The Study tab covers the Mushaf page. You recite from memory first, then reveal line by line to check.

While the curtain is down, the translations, tafsir and the video lesson are hidden too, so nothing can prompt you — and nothing is ever marked merely “read”.

Roediger & Karpicke (2006), *Psychological Science*, 17(3), 249-255 — Experiment 2 final free-recall percentages.

Principle 3 — intervals that expand

- Landauer & Bjork (1978) showed that reviews at expanding gaps beat reviews at equal gaps for learning names — the origin of every expanding-interval system since.
- Later work qualified it: at final test, expanding and equal schedules often perform about the same (Karpicke & Roediger, 2007; Storm, Bjork & Storm, 2010).
- Kang, Lindsey, Mozer & Pashler (2014) explain why an expanding schedule is still preferable in practice.

Average recall DURING training

Expanding • 51%

Equal gaps • 43%

Why that matters for ḥifẓ

A ḥāfiẓ is not preparing for one exam. Every recitation — in ṣalāh, in tarāwīḥ, to a teacher — is itself the performance.

A schedule that keeps recall higher throughout the months of training is therefore better, even when the final-test difference is nil.

An expanding schedule also fails cheaply: an early short gap catches a page that has not stuck. Hence the ladder.

Principle 4 — difficulty that helps

Bjork & Bjork (2011): conditions that slow learning down and feel harder often produce better long-term retention. Fluency is a poor guide to whether something has actually been learned.

Recall before you check

The curtain hides the page. Reciting first is harder than reading along — and that is the point.

Wait until it is nearly forgotten

Intervals are stretched until recall is effortful. A page reviewed while still fresh teaches little.

Rate honestly, not kindly

The 0–5 scale asks how the recitation actually went. A generous rating buys a longer gap you have not earned.

Look-alike verses are surfaced

Mutashābihāt are shown deliberately, because confusable items are where later ḥifẓ breaks.

The consensus ranking of study techniques

Dunlosky, Rawson, Marsh, Nathan & Willingham (2013) graded ten common study techniques on the strength of the evidence behind them. Only two earned the top grade — and the app is built on both.

HIGH UTILITY

Practice testing

Recalling from memory, with feedback

Distributed practice

Spreading study over time

Hifz Companion is these two, applied to one page of the Mushaf at a time.

LOW UTILITY

Re-reading

Passing your eyes over it again

Highlighting

Marking the text

Summarisation

Writing it in your own words

Keyword mnemonics

Imagery tricks for verbal material

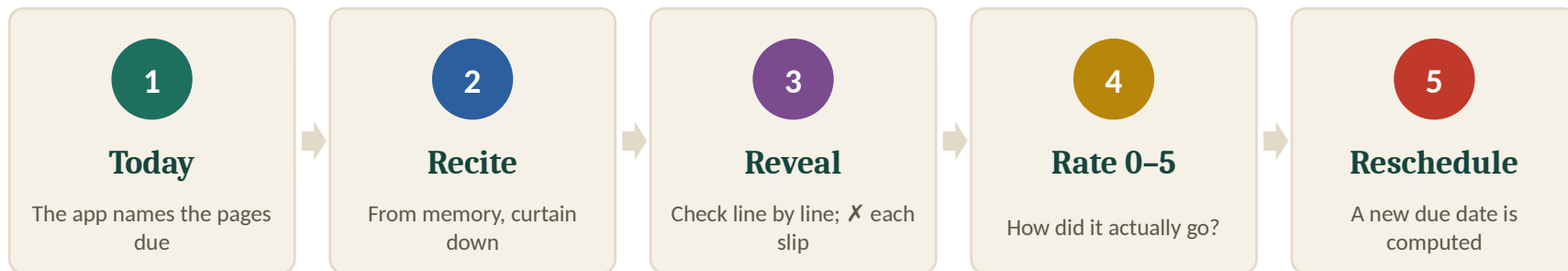
Re-reading is the default behaviour of almost every Qur'an app. It is graded low.

From evidence to interface

- **Forgetting is fastest at first**
 - The first review is scheduled 1 day after a page is completed
- **Optimal gap grows with retention interval**
 - Intervals climb 1 → 3 → 7 → 14 → 30, then multiply
- **Retrieval beats re-reading**
 - The curtain; nothing is ever marked merely “read”
- **Feedback matters**
 - Reveal line by line; ✗ marks the exact āyah that slipped
- **Expanding schedules keep recall high in training**
 - The ladder, rather than a fixed weekly cycle
- **Confusable items break memory**
 - Mutashābihāt shown for each page
- **Difficulty is desirable, fluency is deceptive**
 - A 0–5 honesty rating, not a tick-box

Everything below this line is mechanism: exactly what the scheduler does with a rating.

The daily loop



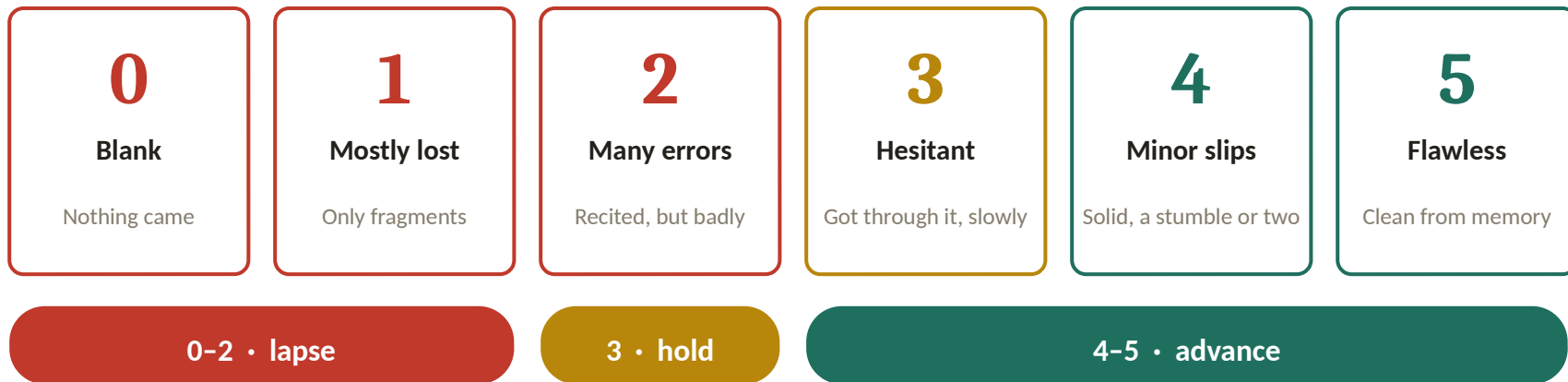
↺ the page returns on its new due date

The one thing that drives everything

Step 4 is the whole system. The rating is the only input the scheduler has, and every interval, priority and promotion downstream is computed from it. No timer and no AI judgement ever substitutes for it.

The rating scale

After each recitation you rate it from 0 to 5. The wording is deliberately about the recitation, not about you.



Three bands, three behaviours. Everything on the following slides is one of these three branches.

A 3 is not a failure — it is the app being told the page is holding but has not grown stronger, so the same interval repeats.

The ladder: 1 · 3 · 7 · 14 · 30

A page that has just been memorised enters the ladder. Each rung is earned by a good recitation (4 or 5) — never by the calendar alone. A page that is not recited well simply does not climb.



rating 4 or 5 at each step

Why these numbers

They approximate the expanding schedules used in the spacing literature and in SuperMemo's early intervals, while staying memorable and roughly weekly-compatible — a ḥāfiẓ can see at a glance why a page is due.

What happens past 30 days

The ladder ends. From there the interval is multiplied by the page's own ease factor, so two pages learned the same week drift apart according to how each one actually behaves for you.

Ease: the page's own multiplier

Every page carries an ease factor. Past the top of the ladder, the next interval is the current interval times ease.

$$\text{next interval} = \text{round}(\text{interval} \times \text{ease})$$

1.3

2.3

2.8

floor

start

ceiling



Rating 5 • flawless

ease + 0.10



Rating 4 • minor slips

ease unchanged



Rating 3 • hesitant

ease - 0.15



Rating 0-2 • lapse

ease - 0.30

What ease is for

Two ḥuffāz, or two pages, do not behave alike. A page of long, rhythmic āyāt may hold for months; a page of dense legal text with look-alike verses may not.

Ease lets the schedule discover that from the ratings instead of assuming it. The clamp stops one bad week, or one lucky streak, running away with the schedule.

Ease factors with a floor of 1.3 follow the SM-2 family (Woźniak & Gorzelańczyk, 1994); the specific step sizes here are the app's own.

What a rating actually does

0, 1 or 2 — lapse

- interval $\rightarrow$ 1 day
- back to the bottom rung
- ease - 0.30
- the page is flagged weak
- lapse counter + 1
- a manzil page is demoted back to sabqi

The page returns tomorrow.

3 — hold

- the same interval repeats
- ease - 0.15
- the good-run counter resets to zero

No climb, no punishment. The page is holding, but it has not shown it is stronger, so it is not given a longer gap.

4 or 5 — advance

- good-run counter + 1
- a 5 adds 0.10 to ease
- climb one rung, or multiply by ease past the top
- two good runs clear the weak flag
- three good runs at ≥ 30 days promote the page to manzil

These are the complete rules. There is nothing else in the scheduler — no hidden weighting, no model fitted to your data, no AI judgement of your recitation.

A slip is remembered, not just punished

Resetting the interval is not enough. A page that has failed once is likely to fail again, so it is marked.

Rating 0-2

weak = true



Next review

one good recitation is not enough —
the flag stays



The one after

two consecutive good recitations clear
it

What the flag buys you

Weak pages are placed first in the day's revision queue, ahead of everything else that is due, and they are offered first in the next week's plan proposal. A page you are losing is worked on while you still have the energy for it.

Why two, not one

One good recitation the day after a failure often reflects the failure itself still being fresh in mind. Requiring two consecutive good recitations, days apart, is a far better signal that the page has actually been repaired.

Graduation into manzil

A page is not moved to the long rotation because time has passed. It has to earn the move.

Both conditions, together

1. three consecutive good recitations (4 or 5)
2. the interval has reached 30 days or more



manzil

steady 30-day rotation, revisited for life

- Graduation is not permanent. A single lapse on a manzil page sends it back to sabqi, where it is worked at short intervals until it is firm again.
- Pages the user marks as “already known” at setup enter manzil directly, but are still shown once soon afterwards rather than trusted blindly.
- The rotation length is a setting. Where the daily cap makes the configured rotation impossible, the app reports the longer realistic figure and labels it an estimate rather than quietly missing the target.

The day has a ceiling

Any spaced-repetition system will eventually generate more work than a person can do. The usual failure is a backlog of hundreds of items that the learner abandons. The app refuses to produce one.

1 • A hard cap

Six revision pages a day by default, and it is a setting.

Whatever the queue length, the day's list is never longer than the cap. Sabaq — the new page — is separate and always shown.

2 • An explicit priority order

When more is due than fits, the queue is sorted in this exact order:

- sabqi before manzil
- weak pages first
- then pages with recent marked mistakes
- then the most overdue
- then by page number

3 • An honest overflow

What does not fit is not dropped and never silently marked done.

It is spread over the following days, counting the work already booked on each of them — so the plan cannot promise a day that is secretly double-booked.

A deliberate ordered comparison, not a weighted score: one page that is 400 days overdue must never outrank a weak page that is due today.

One assessment per page per day

The ladder models memory across days. Rating the same page four times in one evening is not four days of evidence — so the app will not let it become four rungs of interval.



Morning

The page is rated 5. Its state before the rating is snapshotted.



Afternoon

Rated 5 again. The app rewinds to the snapshot and re-applies — the result is one 5, not two.



Evening

Rated 3, on reflection. Rewind, re-apply: the day now counts as a 3. The earlier ratings are corrections, not extra evidence.



Undo

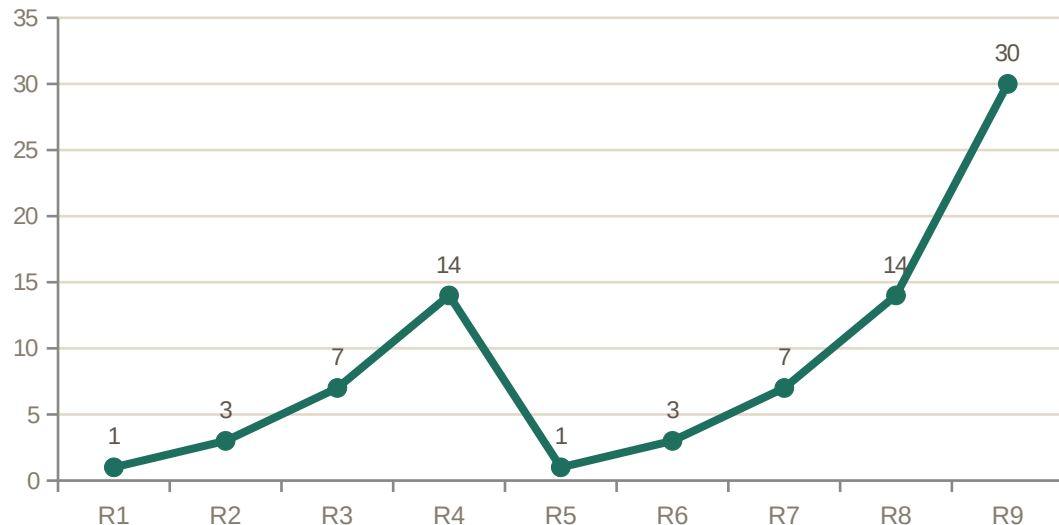
Restores the page exactly as it was this morning, so a mistaken tap can never cost an interval.

Every attempt is still counted and still recorded in the review history — it is only the interval arithmetic that is limited to one assessment a day.

A WORKED EXAMPLE

One page, four months

Page 604, completed on day 0, with a lapse at the fourth review. The line is the scheduled gap in days.



- R1–R4 rated 4 or 5: the page climbs 1 → 3 → 7 → 14.
- R5 rated 1: interval collapses to a single day, ease drops from 2.3 to 2.0, the page is flagged weak and jumps to the front of the queue.
- R6 rated 4: good, but the flag stays.
- R7 rated 5: two consecutive good — the flag clears.
- R9 reaches 30 days with three good in a row, so the page is promoted to manzil and enters the rotation.

A recovering page is not put back where it was; it re-climbs. That is the cost of the lapse, and it is visible to the user in the Progress tab.

DESIGN CHOICES

How it differs from a flashcard scheduler

Aspect	CLASSIC SM-2 FAMILY	HIFZ COMPANION
Unit	A card, a word, a fact	One printed page of the Madinah Mushaf, in the layout you hold
Growth	Ease from the first review; intervals grow fast	A fixed ladder first, ease only past 30 days — slower, more predictable
Ease band	Typically 1.3 upward, unbounded	Clamped 1.3 – 2.8, so no page runs away
Daily load	Whatever is due	Hard cap, explicit priority order, overflow spread honestly
Same-day repeats	Often compound	One assessment per page per day, with undo
Layers	One pool	Three: sabaq, sabqi, manzil — with promotion and demotion

SM-2: Woźniak & Gorzelańczyk (1994). The differences all pull the same way — toward a schedule a human being can sustain and predict.

VERIFICATION

The scheduler is testable, and tested

The scheduling logic is a pure module: no clock, no network, no interface. Every function takes today's date as an argument, so any day can be simulated.

22

scheduler self-tests, runnable from inside
the app

49

browser test suites over the whole
application

0

external services required for any of it

What the self-tests assert

That five consecutive 5s produce exactly 3, 7, 14, 30; that a 1 resets to one day and flags weak; that the flag clears only after two good recitations; that a 3 repeats the interval; that the cap holds.

Why it is written this way

Because a learner's years of work are in this arithmetic. A scheduler that cannot be re-run over a simulated decade, in a second, cannot honestly be trusted with it.

What this app does not claim



No efficacy trial of this app

The design rests on published research about memory in general. The app itself has not been tested in a controlled study against any other method of ḥifẓ, and no such claim is made anywhere in it.



The spacing evidence is general, not Qur'anic

The studies cited used word pairs, passages, facts and faces — not memorisation of a sacred text recited aloud in a fixed rhythm. The principles are well established; their exact quantities are not transferable.



The intervals are a reasoned choice, not a proof

1-3-7-14-30 is defensible and consistent with the literature. It is not derived from an optimisation, and other schedules would also be defensible.



It does not replace a teacher

The app schedules and records. It cannot hear your recitation, and it makes no judgement of tajwīd. Nothing in it substitutes for reciting to a qualified teacher.

SOURCES

References

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Terminology of the three layers: madrasa pedagogical usage, transmitted through teachers; no single canonical text fixes it.

In one sentence

**Recite from memory, say honestly how it went,
and let the schedule decide when the page
returns.**

The three layers are the tradition. The arithmetic between them is spaced repetition.
The app is only the part that remembers.

The app

mazhar17.github.io/hc

User Guide

[.../Hifz-Companion-User-Guide.html](#)

Tutorial (PDF)

[.../Hifz-Companion-Tutorial.pdf](#)

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